

Fibonacci 數列 (佔分 20 分)

問題描述：

費氏數列(Fibonacci Sequence)的前 2 筆資料為 0 和 1，之後的每 1 筆資料為其前 2 筆資料之和，例如第 3 筆為第 1 和 2 筆之和($1+1=2$)，第 4 筆為第 2 和 3 筆之和($1+2=3$)。下表為費氏數列的前 10 筆資料。

i	0	1	2	3	4	5	6	7	8	9
Fib(i)	0	1	1	2	3	5	8	13	21	34

試撰寫一程式印出兩個整數 m 和 n 之間的費氏數字。

輸入說明：

每 1 行讀取 2 個整數 m 和 n，直到讀取 -1 代表輸入資料結束。其中 $1 \leq m < n \leq 100000000$ 。

輸出說明：

根據每行的 m 和 n，輸出兩者之間的所有費氏數字，彼此以空白為間隔；若無費氏數字則輸出 null。

範例 1：

輸入：

4 10

-1

輸出：

5 8

範例 2：

輸入：

100 1000

400 500

200 300

-1

輸出：

144 233 377 610 987

null

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Fibonacci Sequence (20 points)

Problem Description:

The first two numbers of the Fibonacci Sequence are 0 and 1. Every subsequent number is the sum of the two preceding ones. For example, the 3rd number is the sum of the 1st and 2nd ($1+1=2$), and the 4th is the sum of the 2nd and 3rd ($1+2=3$). The following table shows the first 10 numbers of the Fibonacci sequence:

i	0	1	2	3	4	5	6	7	8	9
Fib(i)	0	1	1	2	3	5	8	13	21	34

Write a program to print all Fibonacci numbers between two integers m and n.

Input Description:

Each line reads two integers m and n until -1 is encountered, signifying the end of the input data.

Constraints: $1 \leq m < n \leq 100000000$.

Output Description:

Based on each pair of m and n, output all Fibonacci numbers within that range (inclusive), separated by spaces. If there are no Fibonacci numbers in the range, output null.

Example 1:

Input:

4 10

-1

Output:

5 8

Example 2:

Input:

100 1000

400 500

200 300

-1

Output:

144 233 377 610 987

null

233